

An HQP-Based Obstacle Avoidance Control Scheme for Redundant Mobile Manipulators Under Multiple Constraints

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I. INTRODUCTION

Abstract—It is a challenging problem for redundant wheeled mobile manipulators to satisfy multiple physical constraints and avoid obstacles while performing certain tasks, not to mention constraining tracking performance. In this article, a control scheme based on hierarchical quadratic programming-prescribed performance control (HQP-PPC) is proposed. The first layer of HQP can achieve trajectory tracking, obstacle avoidance, and physical constraints. The second layer of HQP is used to guide the direction of obstacle avoidance and produce a smoother planned path. The prescribed performance function (PPF) can ensure tracking performance so that the physical constraints and obstacle avoidance constraints are satisfied in actual motion. Moreover, the proposed controller can guarantee the tracking performance of the end effector due to the characteristics of the PPF. Finally, the effectiveness and superiority of the proposed method are verified by simulations and experiments.

Index Terms—Hierarchical quadratic program, prescribed performance control (PPC), redundant robot, wheeled mobile manipulators.

A WHEELED mobile manipulator robot (WMMR) can be described as an n -degree of freedom (DOF) manipulator mounted on a wheeled mobile robot. Since WMMRs have a large working space and high DOF, they are widely used in intelligent manufacturing [1], rescue [2], cleaning [3], and nursing [4]. Much research has been done on the control methods of WMMRs in configuration space, such as model-based control [5], adaptive control [6], [7], sliding-mode control [8], and neural network control [9], [10]. However, most WMMR tasks are usually performed by the end effector; therefore, the study of control methods in the task space is more significant.

From the perspective of the dynamic level, the dynamic coupling, nonlinearity, nonholonomic constraints, and uncertainties between the platform and its mounted robot manipulator make tracking performance difficult to guarantee. Therefore, the controllers based on the dynamic model focus more on the estimation and compensation of the unknown model. In [11], Galicki proposed an integral controller in which the manipulator coordinates and the task error plus the gradient of a kinematic criterion function to be integrated to compensate for the errors caused by the uncertainty of the model. In [12], a robust impedance controller based on time-delay compensation for uncertain mobile manipulators was proposed to suppress the unknown disturbances. On the other hand, some researchers have considered the redundancy of WMMRs and used the null space and Moore–Penrose inverse of the Jacobian matrix to solve the obstacle avoidance problem of mobile platforms when WMMRs perform the main tasks [13], [14]. Furthermore, Zhao et al. [15] exploited the redundancy of wheeled humanoid dual-arm robots and achieved holonomic constraints by solving quadratic programming (QP). However, the aforementioned controllers require additional methods to estimate the unknown model or compensate for the disturbances, which makes the controller too complex. In addition, these methods cannot guarantee tracking accuracy and do not address the physical constraints (position and velocity constraints) and obstacle avoidance problem simultaneously.

Determining solutions that better utilize the redundancy of WMMRs has been the subject of numerous studies at the kinematic level. These include achieving repeatable movements [16], [17], physical constraints [18] (joint position and velocity), and

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obstacle avoidance [4]. To the best of the authors' knowledge, there are three main solutions, which are as follows.

- 1) Use the general weighted least-norm (GWLN) inverse and null space to realize constraints [19].
- 2) Solve the QP problem to achieve motion planning [20].
- 3) Use reinforcement learning to obtain the optimal solution for inverse kinematics [21].

In general weighted inverse kinematics, there are inevitably singular problems [22]. The linear variational inequalities-based primal-dual neural network (LVI-PDNN) can solve the QP problem quickly [23], but it may not converge to an optimal solution at some time points and cannot consider subsequent movements, leading to tracking and obstacle avoidance conflicts, such as making the mobile platform stop for obstacle avoidance, which may lead to failure of tracking. Reinforcement learning takes considerable time, especially when the DOF of the robot increases. Due to the updating of motion experiences, it is difficult for reinforcement learning to obtain sufficiently accurate solutions through limited exploration [24]. On the other hand, the aforementioned methods use velocity as the control input, but they ignore the fact that the real-time velocity of the motor is not necessarily the same as the control input velocity [22], which may cause the robot to violate constraints on the actual motion even if the planned path is optimal. Furthermore, since velocity tracking is not considered, nondifferentiable functions are used in the planner's design process [4], [20], resulting in an unsmooth planned reference trajectory and increasing the risk of constraint violation in practice.

Overcoming the aforementioned drawbacks in a single controller or planner is very difficult. To suppress and predetermine the velocity tracking error, a dynamic model must be considered because this is the only way to constrain this error. Most existing online planning methods rely on accurate model information [13]–[20], [22], [23]; however, the dynamic model information is difficult to measure accurately, leading to unsatisfactory planning results. Therefore, a planner-controller system should be organized. The planner should be designed at the kinematic level and must consider the velocity tracking error. The controller must execute the planned reference trajectory with high precision and determine the errors bound in advance.

HQP can realize subtasks as much as possible while executing the main tasks [25]. This property of classifying tasks makes it possible to better utilize the redundancy of robots and complete multiple tasks with low computational cost [26]. Because of this property, HQP is widely used in humanoid robots [27]. The aforementioned planning problems can be better solved by adding subtasks that specify the direction of obstacle avoidance and by considering tracking errors. For the control methods, there are studies on high-precision control methods, such as BLF-based control methods [28], [29] and error transformation [30]. The model-free prescribed performance control (PPC) method has been widely used for robotic arms [31], mobile robots [32], and multiple-input–multiple-output systems [33] due to its insensitivity to model information and concise form. However, this method has not been used to track the trajectory of redundant mobile manipulator end effectors due to their nonholonomic constraints and high degrees of freedom.

To summarize, an HQP-PPC scheme is proposed to avoid obstacles without violating the physical constraints and tracking performance constraints. Compared to the aforementioned methods [4], [22], the HQP-PPC scheme has the following advantages.

- 1) HQP enables the subtask of providing obstacle avoidance directions. In addition, the obstacle avoidance function is optimized. These improvements can provide a smoother reference trajectory, increase the probability of successful obstacle avoidance, and simplify the design of a controller.
- 2) By using prescribed performance function (PPF) to constrain the tracking performance, the bound of velocity tracking error can be predetermined and used in the design of HQP to ensure that the constraints are not violated in actual motion.
- 3) In the HQP-PPC scheme, HQP fully exploits the redundancy of mobile manipulators and provides a smooth reference trajectory for PPC. PPC takes advantage of its ability to set the bound of tracking error in advance to ensure accurate tracking of the reference trajectory.
- 4) The proposed control scheme can ensure tracking performance without violating the constraints and realize obstacle avoidance.

The rest of this article is organized as follows. In Section II, the WMR system is described. The controller design for WMR is derived in Section III, including the introduction of PPF, path planning, and control law design. Simulations and experiments are designed in Sections IV and V, respectively. Section VI discusses the defects of the proposed method and corresponding solutions, and other possible methods for this problem are probed. Finally, Section VII concludes this article.

II. PRELIMINARIES AND PROBLEM FORMULATION

A. List of Symbols

In Table I, the main symbols used across this paper are reported.

B. Kinematics Modeling

A WMMR can be described as an m_a DOF manipulator mounted on a wheeled mobile robot. The end-effector coordinates $O_e = [X, Y, Z]^T$ of the WMMR can be expressed as a function of the joint angles $q_a = [q_{a1}, q_{a2}, \dots, q_{am_a}] \in R^{m_a}$, and the coordinates of mobile platform $q_v = [x_v, y_v, \theta_v]^T$. Let $q = [q_v^T, q_a^T]^T \in R^m$, we have

$$O_e = f_1(q) \quad (1)$$

Then, the end-effector velocity is

$$\dot{O}_e = J_1 \dot{q} \quad (2)$$

where $J_1 = \partial f_1 / \partial q \in R^{3 \times m}$. For the WMMR, there is a full-rank matrix $H_v(q_v) = [\cos \theta_v \ \sin \theta_v \ 0; 0 \ 0 \ 1]^T$ such that we have

$$\dot{q}_v = H_v(q_v) \dot{\eta} \quad (3)$$

where $\dot{\eta} = [v, w]^T$, v represents the line velocity and w represents the angular velocity of the mobile platform in the robot's local coordinate frame. Define $\dot{\xi} = [\dot{\eta}^T, \dot{q}_a^T]^T \in R^n$; then,

$$\dot{q} = H(q)\dot{\xi} \quad (4)$$

where $H(q) = [H_v \ 0; 0 \ I] \in R^{m \times n}$. Substituting (4) into (2), we can obtain the relation between the velocity of the end effector, manipulator joint, and mobile platform [11]

$$\dot{O}_e = J_2 \dot{\xi} \quad (5)$$

where $J_2 = J_1 H(q_v) \in R^{3 \times n}$.

C. Dynamics Modeling

The dynamics of the WMMR can be expressed as the following form [34]:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = B(q)\tau + A(q)^T \chi \quad (6)$$

where $M(q) \in R^{m \times m}$ denotes the inertia matrix, which is symmetric bounded positive definite, $C(q, \dot{q})\dot{q} \in R^m$ is the centripetal and Coriolis torque vector, $G(q) \in R^m$ expresses the gravitational torque vector, $B(q) \in R^{m \times n}$ is a full-rank input transformation matrix, $A(q) \in R^{2 \times m}$ is the full-rank matrix, and χ denotes the Lagrangian multiplier with the nonholonomic constraints. $\tau \in R^n$ is the torque generated by the driving motor.

The WMMR can be separated into two parts: the mobile platform and the manipulator. Therefore, the dynamics of the WMMR (6) can be written as [6]

$$\begin{aligned} M(q) &= \begin{bmatrix} M_v & M_{va} \\ M_{av} & M_a \end{bmatrix}, C(q, \dot{q}) = \begin{bmatrix} C_v & C_{va} \\ C_{av} & C_a \end{bmatrix} \\ G(q) &= \begin{bmatrix} G_v \\ G_a \end{bmatrix}, B(q) = \begin{bmatrix} B_v & 0 \\ 0 & B_a \end{bmatrix} \\ \tau &= \begin{bmatrix} \tau_v \\ \tau_a \end{bmatrix}, \chi = \begin{bmatrix} \chi_v \\ 0 \end{bmatrix}, A = \begin{bmatrix} A_v & 0 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

where $A_v(q_v) \in R^{2 \times 3}$ is the full-rank kinematic constraint matrix. From [6], we have $H_v^T(q_v)A_v^T(q_v) = 0$, and the matrix $H_v^T H_v$ is also full rank.

According to (4) and its derivative, the dynamics of the WMMR can be expressed in the following form [7]:

$$M_1(\xi)\ddot{\xi} + C_1(\xi, \dot{\xi})\dot{\xi} + G_1(\xi) = u_1 \quad (7)$$

where $M_1 = [H_v^T M_v H_v, H_v^T M_{va}; M_{av} H_v, M_a] \in R^{n \times n}$, $C_1 = [H_v^T M_v \dot{H} + H_v^T C_v H_v, H_v^T C_{va}; M_{av} \dot{H} + C_{av} H_v, C_a] \in R^{n \times n}$, $G_1 = [H_v^T G_v; G_a] \in R^n$, and $u_1 = [H_v^T B_v \tau_v; B_a \tau_a] \in R^n$.

D. Prescribed Performance Function (PPF)

To ensure the tracking performance of the end effector and avoid violating the constraints, sufficient tracking accuracy for the reference trajectory is required. Therefore, this section summarizes the preliminary knowledge necessary for the PPC.

As shown in Fig. 2, The prescribed performance is achieved if each element $e_i(t)$, $i = 1, \dots, n$ of the tracking error $e(t) \in R^n$

satisfies the following inequalities for $\forall t \geq 0$:

$$-\delta_i \rho_i(t) < e_i(t) < \rho_i(t) \quad \text{in case of } e_i(0) \geq 0 \quad (8)$$

$$-\rho_i(t) < e_i(t) < \delta_i \rho_i(t) \quad \text{in case of } e_i(0) \leq 0 \quad (9)$$

where $\delta_i(t) \in (0, 1]$, $i = 1, 2, \dots, n$ are used to denote the overshoot allowed for error $e_i(t)$, and $\rho_i(t)$, $i = 1, 2, \dots, n$ are bounded, smooth, strictly positive, and decreasing functions of time, which satisfy $\lim_{t \rightarrow +\infty} \rho_i(t) > 0$, $i = 1, 2, \dots, n$. In this article, the function $\rho_i(t)$ is defined as [35]

$$\rho_i(t) = (\rho_{0i} - \rho_{\infty i}) \varpi_i^{-1} + \rho_{\infty i} \quad (10)$$

where ρ_{0i} and $\rho_{\infty i}$ are positive constants and ρ_{0i} should satisfy $\rho_{0i} > |e_i(0)|$. ϖ_i is a rate function that has the following form [35]:

$$\varpi_i = \begin{cases} \left(\frac{T_i}{T_i - t} \right)^4 e^{k_{ti} t}, & 0 \leq t < T_i \\ \infty, & t \geq T_i \end{cases} \quad (11)$$

k_{ti} is a positive constant that can be designed. $\varpi_i(0) = 1$, where $0 < T_i < \infty$ is a finite time specified by the user. To ensure that ϖ_i is continuous everywhere, we should define $\lim_{t \rightarrow T^-} \left(\frac{T_i}{T_i - t} \right)^4 e^{k_{ti} t} = \infty$.

To satisfy the prescribed performance for tracking error $e_i(t)$, we define the error transformation

$$\varepsilon_i(t) = T_i(\hat{e}_i(t)) \quad (12)$$

$\hat{e}_i(t) \triangleq \frac{e_i(t)}{\rho_i(t)}$ is the normalized tracking error, and the transformation function $T_i(\cdot)$, $i = 1, \dots, n$ is a smooth, strictly increasing function proposed by [31] as follows

$$T_i(\hat{e}_i) = \ln \left[\frac{\delta_i + \hat{e}_i}{\delta_i(1 - \hat{e}_i)} \right] \quad \text{in case of } e_i(0) \geq 0 \quad (13)$$

$$T_i(\hat{e}_i) = \ln \left[\frac{\delta_i(1 + \hat{e}_i)}{1 - \hat{e}_i} \right] \quad \text{in case of } e_i(0) \leq 0. \quad (14)$$

For the convenience of the control law design, the derivative of ε_i is

$$\dot{\varepsilon}_i(t) = \lambda_i \rho_i^{-1}(t) [\dot{e}_i(t) + \alpha_i e_i(t)] \quad (15)$$

where λ_i and α_i are defined as

$$\lambda_i \triangleq \frac{\partial T_i}{\partial \hat{e}_i} > 0, \alpha_i \triangleq -\frac{\dot{\rho}_i}{\rho_i(t)} \geq 0. \quad (16)$$

It is easy to verify that there is a positive constant ϑ_i satisfies [33]

$$e_i \lambda_i \varepsilon_i = \vartheta_i |e_i|^2. \quad (17)$$

The satisfaction of the prescribed performance boundary of tracking error e is directly related to the boundedness of transformed error ε . In fact, given $|\varepsilon_i| \leq \bar{\varepsilon}_i$, for the positive constant $\bar{\varepsilon}_i$, the normalized tracking error $\hat{e}_i$ is confined as follows:

$$\underline{\Omega}_i = \{\hat{e}_i \in R : -\underline{\delta}_i \leq \hat{e}_i \leq \bar{\delta}_i\} \quad \text{in case of } e_i(0) \geq 0 \quad (18)$$

$$\underline{\Omega}_i = \{\hat{e}_i \in R : -\bar{\delta}_i \leq \hat{e}_i \leq \underline{\delta}_i\} \quad \text{in case of } e_i(0) \leq 0 \quad (19)$$

where $\underline{\delta}_i$ and $\bar{\delta}_i$ are constants satisfying $\delta_i < \underline{\delta}_i < \bar{\delta}_i < 1$. Hence, the prescribed performance problem of the guaranteed inequality constraints (8) and (9) are transformed into the bounded problem of the transformed error ε_i .

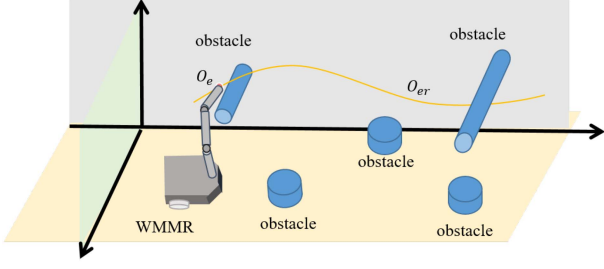


Fig. 1. WMMR tracks the trajectory of the end effector in a multiobstacle environment.

Before designing the control scheme, some assumptions and lemmas need to be presented.

Assumption 1: The desired trajectory O_{er} is bounded and once differentiable.

Assumption 2: All obstacles can be represented by a point with a radius or by a line segment with a radius. Assuming that the WMMR consists of line segments, the distance between the WMMR and an obstacle can be determined by finding the distance from point to line or from line to line. The obstacle avoidance threshold is determined by the radius of the joints or mobile platform and the radius of the obstacle.

Lemma 1: [31] If $f(t, x)$ and $[\partial f / \partial x](t, x)$ are continuous on $[a, b] \times D$, for some domain $D \subset R^n$, then f is locally Lipschitz in x on $[a, b] \times D$, and there exists a maximal solution $x(t)$ for the initial value problem (IVP)

$$\dot{x} = f(t, x), \quad x(0) = x_0 \in D$$

on a time interval $[0, \tau_{\max})$ with $\tau_{\max} \in \{R_+^*, \infty\}$, such that $x(t) \in D \forall t \in [0, \tau_{\max})$.

Lemma 2: [33] Consider an IVP and assume that the hypotheses of Lemma 1 are upheld. For a maximal solution $x(t)$ defined on the interval $[0, \tau_{\max})$, if $\tau_{\max} \neq \infty$, then for any compact set $D' \subset D$, there exists a time instant $\tau' \in [0, \tau_{\max})$ such that $x(\tau') \notin D'$.

III. CONTROLLER DESIGN

As shown in Fig. 1, The control objective is to design a controller for the WMMR so that it can follow a desired trajectory $O_{er} = [O_{erx}; O_{ery}; O_{erz}]$ within a prescribed performance constraint while ensuring that the WMMR operates in safe areas, such as achieving joint or velocity limits and obstacle avoidance.

The main ideas of this control scheme are as follows: HQP is used to account for physical constraints and obstacle avoidance to obtain reference velocity $\dot{\xi}_r$. Control input u_1 is provided by PPC to ensure tracking accuracy. PPC provides the velocity tracking error boundary for HQP and ensures that the constraints are not violated even if there is a velocity tracking error. HQP provides the reference velocity for PPC, which allows the robot to avoid obstacles while tracking the desired trajectory O_{er} .

A. HQP-Based Path Planning

In this section, we discuss obtaining the reference velocity $\dot{\xi}_r \in R^n$ in the configuration space by HQP. HQP can realize

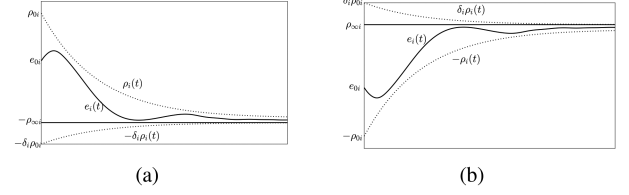


Fig. 2. Constructing performance bounds by performance functions.

path planning according to the priority order of the task. In this article, there are three main goals that must be realized, which are as follows:

- 1) the end trajectory can track the target trajectory;
- 2) obstacle avoidance;
- 3) physical constraints.

The secondary objective is to guide the direction of obstacle avoidance.

The main objectives can be written as the following QP question:

$$\text{minimize } \frac{1}{2}(\dot{\xi}_r - \dot{\xi}_s)^T(\dot{\xi}_r - \dot{\xi}_s) \quad (20)$$

$$\text{subject to } J_2 \dot{\xi}_r = \dot{O}_{er} - K_e \lambda_o \varepsilon_o \quad (21)$$

$$\tilde{J}_1 \dot{\xi}_r \leq \tilde{v} \quad (22)$$

$$\tilde{\zeta}^- \leq \dot{\xi}_r \leq \tilde{\zeta}^+ \quad (23)$$

$\dot{\xi}_s \in R^n$ will be obtained later through the second-level QP problem and used to achieve the subtask. In the first-level QP problem, we can accomplish the secondary objective as much as possible by minimizing (20). The remaining equation and inequalities are related to the main task. These parameters in the aforementioned QP question are detailed as follows.

1) Trajectory Constraint: Equation (21) represents the trajectory constraint of the end effector. From (12), the normalized errors ε_o can be transformed by

$$\varepsilon_{oi} = T_i(\hat{e}_{oi}) \quad i = 1, 2, 3 \quad (24)$$

where $\hat{e}_{oi} = \frac{e_{oi}(t)}{\rho_{oi}(t)}$ and

$$e_o = O_e - O_{er} \quad (25)$$

represent the position tracking error, the gain matrix $K_e = \text{diag}_{i \in \{1,2,3\}}[k_{ei}]$ is positive, and $\lambda_o = \text{diag}_{i \in \{1,2,3\}}[\lambda_{oi}]$ is defined by (16).

2) Obstacle Avoidance Constraint: Inequality (22) can implement the obstacle avoidance constraint. Obstacle avoidance can be seen as the critical point on the robot that must maintain a certain distance from the obstacle point. To achieve this goal, we find a point $p = [x_p, y_p, z_p]^T$ on the robot that is closest to the obstacle $o = [x_o, y_o, z_o]^T$, as shown in Fig. 3, and define the distance d between the obstacle and the robot as

$$d = (p - o)^T(p - o). \quad (26)$$

For distance d , we set two thresholds d_1 and d_2 , where $d_1 > d_2$. Obstacle avoidance should be started when d is less than d_1 , and d_2 is the distance that is not allowed to be reached, which

is usually determined by the radius of the obstacle and the robot joint. To facilitate the design of the constraint condition, we normalized the distance d as

$$d_n = \begin{cases} 1, & d < d_2 \\ \cos^2\left(\frac{\pi(d-d_1)}{2(d_2-d_1)}\right), & d_2 \leq d \leq d_2 \\ 0, & d > d_1. \end{cases} \quad (27)$$

We define the following function $g(d_n)$ with respect to the normalized distance:

$$g(d_n) = \begin{cases} 0, & d_n < 0 \\ 0.1 \tan\left(\frac{d_n \pi}{2}\right), & 0 \leq d_n \leq 1 \\ \infty, & d_n > 1. \end{cases} \quad (28)$$

To complete obstacle avoidance, the function $g(d_n)$ should be monotonically nonincreasing. Therefore, the following inequality needs to be satisfied:

$$\frac{\partial g}{\partial q} \dot{q} \leq 0. \quad (29)$$

Let $\tilde{J}_1 = \frac{\partial g}{\partial q} H \in R^{1 \times n}$, and consider the velocity tracking errors

$$e_\xi = \dot{\xi} - \dot{\xi}_r \quad (30)$$

and their bound $\rho_\xi \in R^n$ caused by PPC, which will be proven in the next section. Then, we have $\tilde{J}_1 \xi_r + |\tilde{J}_1| |\rho_\xi| \geq \tilde{J}_1 \xi = \frac{\partial g}{\partial q} \dot{q}$. Defining $\tilde{v} = -|\tilde{J}_1| |\rho_\xi|$, the obstacle avoidance constraint (22) is obtained. We should note that when $d > d_1$, we have $\frac{\partial g}{\partial q} = 0$, and inequality (22) has no effect on ξ_r .

Remark 1: Compared to [4], [20], and [22], the proposed method does not use *sign* or other nondifferentiable functions, and the obstacle avoidance function $g(d_n)$ is continuously differentiable so that the reference velocity ξ_r is smooth, which is important for the subsequent design of the controller.

3) Joint Limit Constraint: Inequality (23) represents the joint limit constraint. Physical limits are common problems for practical robot systems. It is necessary to consider physical constraints to ensure the safety of the robot. The physical constraints include the limits of joint angles and the limits of robot velocity

$$q_a^- \leq q_a \leq q_a^+, \quad \dot{\xi}^- \leq \dot{\xi} \leq \dot{\xi}^+ \quad (31)$$

where $q_a^\pm \in R^{m_a}$ and $\dot{\xi}^\pm \in R^n$ denote the bounds of the joint angles and velocities of the robot. According to [22], (31) can be converted into the following inequalities:

$$\zeta^- \leq \dot{\xi} \leq \zeta^+ \quad (32)$$

where $\zeta^- = \max\{a(\zeta^- - \zeta), \dot{\xi}^-\}$, $\zeta^+ = \min\{a(\zeta^+ - \zeta), \dot{\xi}^+\}$, $\zeta^- = [-\infty, -\infty, q_a^{-T}]^T$, $\zeta = [0, 0, q_a^T]^T$, $\zeta^+ = [+\infty, +\infty, q_a^{+T}]^T$, and a is a positive constant determined by the control frequency.

Considering velocity tracking errors and their bound ρ_ξ , we have

$$\zeta^- + \rho_\xi \leq \dot{\xi}_r \leq \zeta^+ - \rho_\xi. \quad (33)$$

Letting $\tilde{\zeta}^- = \zeta^- + \rho_\xi$ and $\tilde{\zeta}^+ = \zeta^+ - \rho_\xi$, we can obtain the joint limit constraint (23).

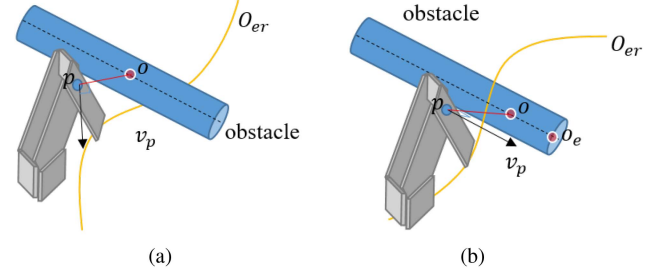


Fig. 3. Direction of obstacle avoidance. (a) When the desired trajectory is below the obstacle. (b) When the desired trajectory is above the obstacle.

Remark 2: The possible error in velocity tracking is taken into account in HQP so that the robot does not violate the constraints in actual operation. In contrast, the error range must be determined in advance, which requires a higher control performance.

4) Guiding the Direction of Obstacle Avoidance: Inequality (22) does not specify the direction of obstacle avoidance, and the obstacle avoidance scheme may be rational at the present time, but the solution obtained is only the optimal solution at the present time and does not take into account the entire trajectory, which may cause obstacle avoidance and trajectory following to potentially conflict and ultimately cause the singularity of the Jacobian matrix (e.g., bringing the mobile platform to a halt, which is very likely with minimization of $\dot{\xi}_r$, especially in the face of the obstacles on the ground).

To overcome this problem and obtain a smoother desired path, the following second-level QP problem is used to guide the direction of obstacle avoidance:

$$\text{minimize } \frac{1}{2} \dot{\xi}_s^T \dot{\xi}_s \quad (34)$$

$$\text{subject to } \tilde{J}_1 \dot{\xi}_s = v_p \quad (35)$$

$v_p \in R^3$ will be determined by the position of the obstacle, the desired trajectory of the end effector O_{er} and the distance from the obstacle. As shown in Fig. 3, when O_{erz} is below the obstacle

$$v_p = [0; 0; -g(d_n)]. \quad (36)$$

When O_{erz} is above the obstacle, we need to find the closest endpoint o_e on the obstacle; then,

$$v_p = g(d_n)[o_{ex} - o_x; o_{ey} - o_y; 0]. \quad (37)$$

The aforementioned real-time online solution methods of QP problems are all referred to in [17]. From [17, Appendix 1], for all $\hat{e}_o \in \underline{\Omega}_o$, we have

$$J_2 \dot{\xi}_r = \dot{O}_{er} - K_e \lambda_o \varepsilon_o + \varepsilon \quad (38)$$

$\dot{\xi}_r$ is bounded, and there is a positive constant $\bar{\varepsilon}$ that satisfies $|\varepsilon| \leq \bar{\varepsilon}$.

B. Control Law Design

Define the following transformed velocity error:

$$\varepsilon_{\xi i} = T_i(\hat{e}_{\xi i}) \quad i = 1, 2, \dots, n \quad (39)$$

where $\hat{e}_{\xi i} = \frac{e_{\xi i}(t)}{\rho_{\xi i}(t)}$. When the errors ε_{oi} and $\varepsilon_{\xi i}$ are bounded, the tracking errors e_o and $\dot{e}_o$ are in the prescribed region.

We propose the following model-free control law:

$$u_1 = -J_2^T K_o \lambda_o \varepsilon_o - K_{\xi} \lambda_{\xi} \varepsilon_{\xi} \quad (40)$$

where $K_o = \text{diag}_{i \in \{1,2,3\}}[k_{oi}]$ and $K_{\xi} = \text{diag}_{i \in \{1,\dots,n\}}[k_{\xi i}]$ are the positive diagonal gain matrices and $\lambda_{\xi} = \text{diag}_{i \in \{1,\dots,n\}}[\lambda_{\xi i}]$ is defined by (16).

Theorem 1: Consider system (7) under *Assumptions 1* and 2, the model-free control law (40) with the overshoot indices $\delta_o, \delta_{\xi} \in (0, 1]$, and the performance functions $\rho_o(t), \rho_{\xi}(t)$, among which $\rho_o(0), \rho_{\xi}(0)$ should be selected to satisfy $\rho_{oi}(0) > |e_{oi}(0)|, i = 1, 2, 3$ and $\rho_{\xi i}(0) > |e_{\xi i}(0)|, i = 1, \dots, n$, respectively, then all closed-loop signals are bounded and the prescribed performance of the tracking errors e_o is guaranteed.

Proof: *Theorem 1* is proven in four steps.

Step 1: Substituting the control law (40) and tracking errors and (30) into the system model (7), we have

$$M_1 \dot{e}_{\xi} + C_1 e_{\xi} + J_2^T K_o \lambda_o \varepsilon_o + K_{\xi} \lambda_{\xi} \varepsilon_{\xi} + \Delta_1 = 0 \quad (41)$$

where $\Delta_1 = M_1(\ddot{\xi}_r - J_2^T K_e e_o - J_2^T K_e \dot{e}_o) + C_1(\dot{\xi}_r - J_2^T K_e e_o) + G_1$. It is easy to ascertain that there is a positive constant $\bar{\Delta} \forall \hat{e}_o \in \underline{\Omega}_o$ and that $\forall \hat{e}_{\xi} \in \underline{\Omega}_{\xi}$ satisfies the inequality $\|\Delta_1\| \leq \bar{\Delta}_1$ from the previous section. The aforementioned state-space error system (41) can be written with respect to the normalized states $\hat{e} = [\hat{e}_o^T \hat{e}_{\xi}^T]^T$ as [31]

$$\dot{\hat{e}} = f(\hat{e}, t) \quad (42)$$

with $f(\hat{e}, t) : \underline{\Omega} \times R^+ \rightarrow R^{n+3}$ with $\underline{\Omega} = \underline{\Omega}_{\xi} \times \underline{\Omega}_o$. By constructing the appropriate $\delta_{oi}, \delta_{\xi i}, \rho_{oi}(0)$ and $\rho_{\xi i}(0)$, we can let $\hat{e}(0) \in \underline{\Omega}$. Therefore, the locally Lipschitz continuity of $f(\hat{e}, t)$ can be concluded invoking *Lemma 1*. Then, the system (42) has a unique maximal solution over a time interval $[0, t_{\max})$, which implies that the transformed errors $\varepsilon_o, \varepsilon_{\xi}$ and λ_o, λ_{ξ} are well defined over $t \in [0, t_{\max})$.

Step 2: Consider the following Lyapunov function:

$$V_1(e_{\xi}, \varepsilon_o, t) = \frac{1}{2} e_{\xi}^T M_1 e_{\xi} + \frac{1}{2} \varepsilon_o^T K_o \rho_o \varepsilon_o. \quad (43)$$

Substituting the system trajectories of (41) into the derivative of V_1 yields

$$\begin{aligned} \dot{V}_1 = & -e_{\xi}^T K_{\xi} \lambda_{\xi} \varepsilon_{\xi} - e_{\xi}^T J_2^T K_o \lambda_o \varepsilon_o \\ & + e_{\xi}^T \Delta_1 + \varepsilon_o^T K_o \rho_o \dot{e}_o + \frac{1}{2} \varepsilon_o^T K_o \dot{\rho}_o \varepsilon_o. \end{aligned} \quad (44)$$

Utilizing $\dot{e}_o \leq J_2 e_{\xi} - K_e \lambda_o \varepsilon_o + \bar{e}$, then $\dot{e}_o \leq \lambda_o \rho_o^{-1} [J_2 e_{\xi} - K_e \lambda_o \varepsilon_o + \bar{e} + \alpha_o e_o]$, where $\alpha_o = \text{diag}_{i \in \{1,\dots,n\}}[\alpha_{oi}]$. Therefore, (44) can be expressed as

$$\begin{aligned} \dot{V}_1 = & -e_{\xi}^T K_{\xi} \lambda_{\xi} \varepsilon_{\xi} - \varepsilon_o^T K_o \lambda_o \lambda_o^T K_e e_o + \varepsilon_o^T K_o \lambda_o \bar{e} \\ & + e_{\xi}^T \Delta_1 + \varepsilon_o^T K_o \lambda_o \alpha_o e_o + \frac{1}{2} \varepsilon_o^T K_o \dot{\rho}_o \varepsilon_o. \end{aligned} \quad (45)$$

Since Δ_1 is bounded, $e_{\xi}^T \Delta_1 \leq \bar{\Delta}_1^2/2 + \|e_{\xi}\|^2/2$. According to (17), $e_{\xi}^T K_{\xi} \lambda_{\xi} \varepsilon_{\xi} = \vartheta_{\xi} \|e_{\xi}\|^2$, where ϑ_{ξ} is a positive constant. In view of Young's inequality [35], we have $\varepsilon_o^T K_o \lambda_o \bar{e} \leq \frac{1}{\sigma_1} K_o^2 \varepsilon_o^T \lambda_o \lambda_o^T \varepsilon_o + \sigma_1 \bar{e}^T \bar{e}$; then, K_o , and K_e can be designed so that there is always a constant σ_1 that

satisfies inequality $-\varepsilon_o^T K_o \lambda_o \lambda_o^T K_e e_o + \frac{1}{\sigma_1} K_o^2 \varepsilon_o^T \lambda_o \lambda_o^T \varepsilon_o + \varepsilon_o^T K_o \lambda_o \alpha_o e_o \leq -\vartheta_o \|\varepsilon_o\|^2$, where ϑ_o is a positive constant.

From the requirement of the performance function, $\dot{\rho}_o < 0 \forall t \in [0, t_{\max})$, then $\frac{1}{2} \varepsilon_o^T K_o \dot{\rho}_o \varepsilon_o < 0$ by the design of K_o . Hence, (45) can be expressed as

$$\begin{aligned} \dot{V}_1 \leq & -\left(\vartheta_{\xi} - \frac{1}{2}\right) \|e_{\xi}\|^2 - \vartheta_o \|\varepsilon_o\|^2 + \frac{1}{2} \bar{\Delta}_1^2 + \sigma_1 \bar{e}^T \bar{e} \\ \leq & -\kappa_1 V_1 + \nu_1 \end{aligned} \quad (46)$$

where $\kappa_1 = \min\{\lambda_{\min}(\vartheta_{\xi} - \frac{1}{2})/\lambda_{\max}(M_1), \vartheta_o/\lambda_{\max}(K_o \rho_o)\}$, and $\nu_1 = \frac{1}{2} \bar{\Delta}_1^2 + \sigma_1 \bar{e}^T \bar{e}$. It is easy to ascertain that $\hat{e}_o \in \underline{\Omega}_o \forall t \in [0, t_{\max})$, and $\underline{\Omega}_o = \{\hat{e}_o \in R : |\varepsilon_o| \leq \sqrt{(2(V_1(0) + (\nu_1/\kappa_1)))}/\lambda_{\max}(K_o \rho_o)\}$ from [29]. For the set $\underline{\Omega}_o$, there is always a constant $\bar{\varepsilon}_o$ large enough for $\underline{\Omega}_o$ to be a compact subset of $\underline{\Omega}_o$.

Step 3: Consider the following Lyapunov function:

$$V_2(\varepsilon_{\xi}) = \frac{1}{2} \varepsilon_{\xi}^T K_{\xi} \varepsilon_{\xi}. \quad (47)$$

The derivative of ε_{ξ} with respect to time is:

$$\dot{\varepsilon}_{\xi} = -\lambda_{\xi} \rho_{\xi}^{-1} M_1^{-1} K_{\xi} \lambda_{\xi} \varepsilon_{\xi} - \lambda_{\xi} \rho_{\xi}^{-1} \Delta_2 \quad (48)$$

where $\Delta_2 = M_1^{-1}(C_1 e_{\xi} + J_3^T K_o \lambda_o \varepsilon_o + \Delta_1) + \alpha_{\xi} e_{\xi}$. Since ε_o has been proven to be bounded $\forall t \in [0, t_{\max})$, Δ_2 satisfies $\|\Delta_2\| \leq \bar{\Delta}_2, \forall t \in [0, t_{\max})$, where $\bar{\Delta}_2$ is a positive constant.

Substituting (48) into the derivative of V_2 yields

$$\dot{V}_2 = -\varepsilon_{\xi}^T K_{\xi} \lambda_{\xi} \rho_{\xi}^{-1} M_1^{-1} K_{\xi} \lambda_{\xi} \varepsilon_{\xi} - \varepsilon_{\xi}^T K_{\xi} \lambda_{\xi} \rho_{\xi}^{-1} \Delta_2. \quad (49)$$

By the properties of the inertia matrix M_1 , there is a $\sigma_2 > 0$ such that $\varepsilon_{\xi}^T K_{\xi} \lambda_{\xi} \rho_{\xi}^{-1} M_1^{-1} K_{\xi} \lambda_{\xi} \varepsilon_{\xi} \leq \sigma_2 \|K_{\xi} \lambda_{\xi} \varepsilon_{\xi}\|^2$. $-\varepsilon_{\xi}^T K_{\xi} \lambda_{\xi} \rho_{\xi}^{-1} \Delta_2 \leq \sigma_3 \|K_{\xi} \lambda_{\xi} \varepsilon_{\xi}\|^2 + \bar{\Delta}_2/(4\sigma_3)$, where σ_3 is an adjusted positive constant to make $\sigma_2 - \sigma_3 > 0$. Hence, (49) can be written as

$$\dot{V}_2 \leq -(\sigma_2 - \sigma_3) \|K_{\xi} \lambda_{\xi} \varepsilon_{\xi}\|^2 + \frac{\bar{\Delta}_2}{4\sigma_3} \leq -\kappa_2 V_2 + \nu_2 \quad (50)$$

where $\kappa_2 = (\sigma_2 - \sigma_3) \|K_{\xi} \lambda_{\xi}\|^2$, $\nu_2 = \frac{\bar{\Delta}_2}{4\sigma_3}$. Then, $\hat{e}_{\xi} \in \underline{\Omega}_{\xi} \forall t \in [0, t_{\max})$, where $\underline{\Omega}_{\xi} = \{\hat{e}_{\xi} \in R : |\varepsilon_{\xi}| \leq \sqrt{(2(V_2(0) + (\nu_2/\kappa_2)))}/\lambda_{\max}(K_{\xi})\}$ from [29] and it is a compact subset of $\underline{\Omega}_{\xi}$ for a sufficiently large $\bar{\varepsilon}_{\xi}$.

Step 4: There is a unique solution $\hat{e}(t)$ for the closed-loop system that satisfies $\hat{e}(t) \in \Omega \subset \underline{\Omega} \forall t \in [0, t_{\max})$. Assuming $t_{\max} < \infty$, according to *Lemma 2*, there exists a time instant $t \in [0, t_{\max})$ such that $\hat{e}(t) \notin \Omega$, which contradicts $\hat{e}(t) \in \Omega \subset \underline{\Omega} \forall t \in [0, t_{\max})$. Thus, $t_{\max} = \infty$.

From the aforementioned proof, the control law (40) can ensure that the transformed errors ε_o and ε_{ξ} are bounded; then, according to the property of the transform function (12), the tracking errors e_o and e_{ξ} must be within the specified range. Therefore, *Theorem 1* has been proven.

Remark 3: The proposed method can ensure the tracking performance of the end-effector trajectory O_r , suppress the planning error ϵ caused by HQP, determine the tracking accuracy of the planning velocity $\dot{\xi}_r$ and avoid violating the constraints.

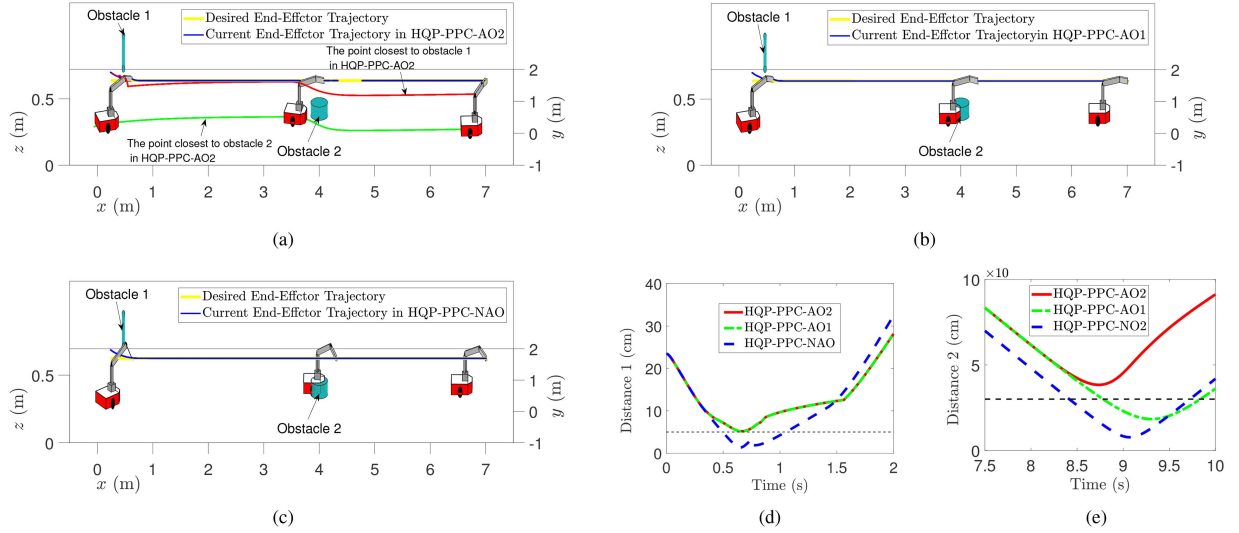


Fig. 4. Simulation results of HQP-PPC. (a) Trajectory of the WMMR in HQP-PPC-AO2. (b) Trajectory of the WMMR in HQP-PPC-AO1. (c) Trajectory of the WMMR in HQP-PPC-NAO. (d) Distance 1 between the robot and obstacle. (e) Distance 2 between the robot and obstacle.

Moreover, the proposed method does not need any model information or fitting model structure, so the complexity of the control scheme is low. ■

IV. SIMULATIONS

To verify the effectiveness of our control scheme, three numerical simulations are carried out. The robot's dynamic model is similar to [34]. The mechanical structure of a three-axis mobile manipulator is shown in Fig. 10(b). The physical parameters of the robots are taken as $r = 0.065$ m and $L_0 = 0.1$ m. The mass of the robot is $m_c = 3$ kg, and the wheel is $m_w = 0.1$ kg. The moment of inertia $I_c = 1.625$ kg · m², $I_w = 0.005$ kg · m², and $I_m = 0.0025$ kg · m² and the gravitational constant $g = 9.8$ m / s². The length and width of the robot chassis are 0.15 m, and the height h is 0.1 m. The length of each joint is $L_1 = 0.2$ m, $L_2 = 0.3$ m, and $L_3 = 0.15$ m, and the radius is 0.025 m. The bounds are $q_a^+ = -q_a^- = [\frac{\pi}{2}, \frac{\pi}{2}, \frac{\pi}{2}]^T$ rad and $\dot{q}^+ = -\dot{q}^- = [2.5, 2.5, 2.5, 2.5, 2.5]^T$ m/s or rad/s. In each case, the simulations have the same initial pose $q_v(0) = [0, 0, \pi/3]^T$, $q_a(0) = [0, \pi/3, -\pi/8]^T$ and desired end-effector trajectory $O_{er} = [0.26 + 0.45t, 0.38, 0.31]^T$. There are two obstacles. The first obstacle is an infinitely long cylinder with a radius of 0.025 m, perpendicular to the x -axis, and its $x = 0.47, z = 0.4$, which is used to demonstrate the effect of the method on the avoidance of obstacles above the moving platform. According to the radius of obstacle 1 and the manipulator, the obstacle avoidance thresholds are $d_1 = 0.05$ m and $d_2 = 0.3$ m. The second obstacle is placed on the ground at $x = 4, y = 0.55, z = 0$, and its radius is 0.15 m. The second obstacle can reflect the obstacle avoidance effect of the moving platform. The obstacle avoidance thresholds are $d_1 = 0.3$ m and $d_2 = 0.6$ m.

A. Test of the Obstacle Avoidance Function

This subsection discusses the effect of HQP-PPC on obstacle avoidance by turning ON and OFF the obstacle avoidance

function. The control gains in HQP-PPC are chosen as $K_e = 0.001$, $K_o = 4$, and $K_\xi = \text{diag}([6, 6, 4, 2, 2])$. The gain of the first QP solver in [17] is 1000, and that of the second QP solver is 500. The parameters of the PPF are $\rho_{0o} = 0.2$, $\rho_{\infty o} = 0.005$, $\rho_{0\xi} = 0.6$, $\rho_{\infty\xi} = 0.1$, $\delta_o = \delta_\xi = 1$, $T_o = T_\xi = 5$, and $k_{to} = k_{t\xi} = 1$.

The simulation cases are as follows.

- 1) *HQP-PPC-AO2*: Turn ON all obstacle avoidance functions to avoid all obstacles.
- 2) *HQP-PPC-AO1*: Turn ON the first obstacle avoidance function to avoid the first obstacle.
- 3) *HQP-PPC-NAO*: Turn OFF all obstacle avoidance functions and not avoid obstacles.

Fig. 4(a)–(c) shows the trajectory of the WMMR by using HQP-PPC. The WMMR will certainly collide with obstacles if the obstacle avoidance function is not performed, which can also be seen from the distance between the obstacles and the critical points in Fig. 4(d) and (e). As shown in Fig. 5, the velocity, position, and tracking error constraints are not violated in HQP-PPC-AO2.

B. Test of the HQP Function

This subsection serves to show the performance of HQP. In this subsection, the two commonly used inverse kinematics methods QP [4] and GWLN [22] are used for comparison. It should be noted that these two methods operate in kinematics, and there are no disturbances in the simulation that affect the performance of these two methods.

1) *Quadratic programming (QP)*: As shown in Fig. 6(a), the QP method is not effective in trajectory tracking of the WMMR with low redundancy. None of the obstacles can be successfully avoided, especially obstacle 2, which directly affects the trajectory tracking of the end effector. This is because the inequality constraint (28) (or [4, inequality (9)]) does not directly provide the force to move away from the obstacle and may cause critical points to stop, which conflicts with the trajectory tracking and

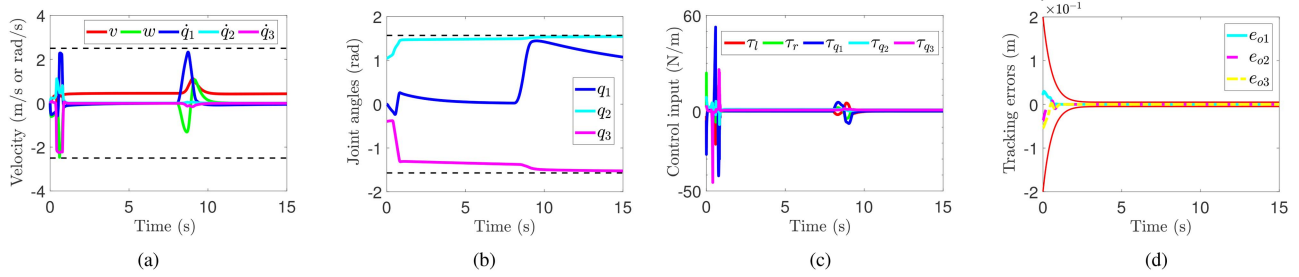


Fig. 5. Simulation results of HQP-PPC-AO2. (a) Velocity of the WMMR. (b) Joint angles of the manipulator. (c) Control input. (d) Tracking errors.

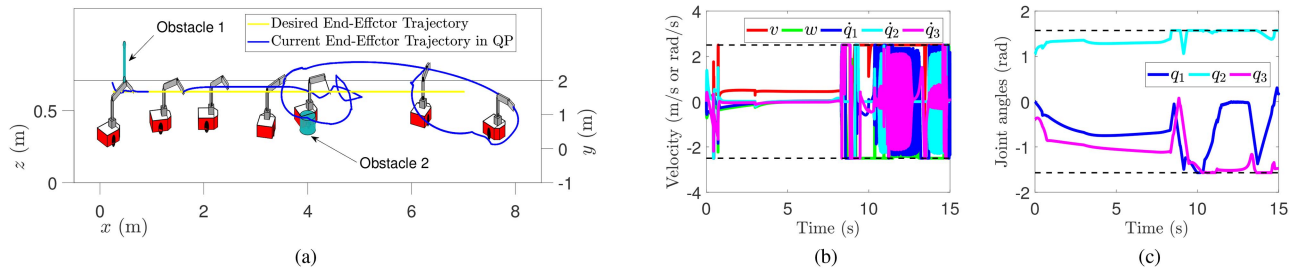


Fig. 6. Simulation results of QP. (a) Trajectory of the WMMR. (b) Velocity of the WMMR. (c) Joint angles of the manipulator.

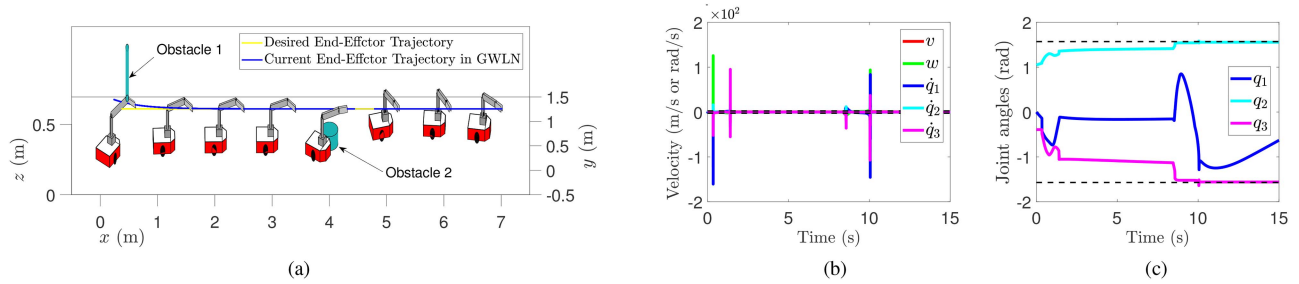


Fig. 7. Simulation results of the GWLN. (a) Trajectory of the WMMR. (b) Velocity of the WMMR. (c) Joint angles of the manipulator.

makes the whole system out of control. On the other hand, the *sign* function makes the velocity discontinuous, as seen in Fig. 6(b); even at relatively stable 0–5 s, the velocity curves are discontinuous, making control at the dynamic level very difficult. In Fig. 6(c), the joint angles of the manipulator are shown. Note that the velocity and position constraints are not violated when the robot is out of control, which is due to the characteristics of LVI-PDNN.

2) *General weighted least-norm (GWLN)*: Fig. 7(a) shows the trajectory of the WMMR using the GWLN method, which shows good performance in tracking. However, due to the shortcomings of the GWLN method, the robot inevitably enters a singularity when it approaches obstacles, which leads to the failure of obstacle avoidance. Due to the singularity, the velocity increases sharply near the obstacles in Fig. 7(b). Therefore, this method must be used with great caution in practical applications. At approximately 10 s in Fig. 7(c), the position is outside of the constraint due to the sudden increase in velocity.

In summary, without HQP, neither single-layer QP nor the GWLN can achieve good obstacle avoidance and obtain a satisfactory solution to inverse kinematics.

C. Test of the PPC Function

This subsection discusses the effect of PPC, in which the two common dynamics control methods, adaptive robust control (ARC) [6] and fuzzy logic control (FLC) [36], are used for comparison.

If the PPF is not applied, not only can the end-effector tracking accuracy shown in Figs. 8(a) and 9(a) not be guaranteed but the velocity constraints shown in Fig. 8(b) and Fig. 9(b) are also violated because the velocity tracking error bound cannot be determined in advance.

V. EXPERIMENTS

To verify the effectiveness of our method, comparative experiments are performed. The experimental apparatus is shown

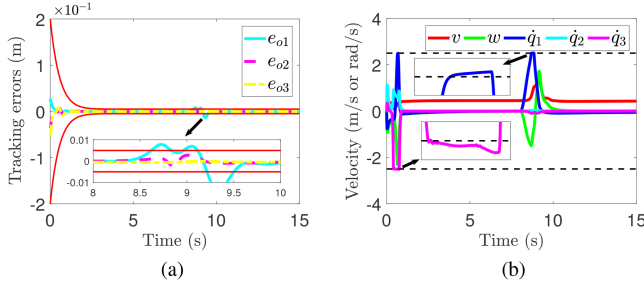


Fig. 8. Simulation results of ARC. (a) Tracking errors. (b) Velocity of the WMMR.

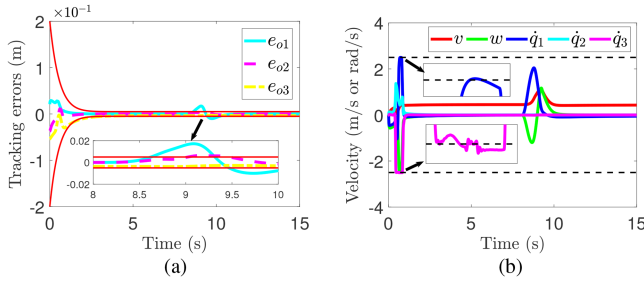


Fig. 9. Simulation results of FLC. (a) Tracking errors. (b) Velocity of the WMMR.

in Fig. 10(a). The host computer obtains the position of the mobile platform via OptiTrack and reads the encoder to obtain the joint angle and velocity of the WMMR. The communication frequency between the host computer and the WMMR is 70 Hz. Since the chatter in the velocity data collected by the sensor affects the control effect, median filtering is performed on all velocity signals.

Since the robot may violate constraints in the comparison experiment, we set $\dot{\xi}^+ = -\dot{\xi}^- = [1, 1, 1, 1, 1]^T$ m/s or rad/s in the experiments for safety. The initial pose is $q_v(0) = [-1.35, 0.65, 0]^T$, $q_a(0) = [0, 0, 0]^T$, and the desired end-effector trajectory is $O_{er} = [0.35 - 3.6 \cos(\pi t/8)/\pi, 0.15 - 3.6 \sin(\pi t/8)/\pi, 0.45 + 0.1 \cos t]^T$. We set up two obstacles. The first obstacle is placed on the ground at $[0.1, -0.85, 0]$, and its radius is 0.15 m. The second obstacle is infinitely long with a radius of 0.025 m, perpendicular to the x -axis, and one end of the stick is $[-0.5, 1.0, 0.35]$. The obstacle avoidance thresholds are $d_1 = 0.05$ m and $d_2 = 0.3$ m.

A. Test of the Obstacle Avoidance Function

The control gains in HQP-PPC are chosen as $K_e = 0.001$, $K_o = 0.1$, and $K_\xi = \text{diag}([0.6, 0.6, 0.4, 0.2, 0.2])$. The parameters of the PPF are $\rho_{0o} = 1.2$, $\rho_{\infty o} = 0.02$, $\rho_{0\xi} = 0.6$, $\rho_{\infty\xi} = 0.1$, $\delta_o = \delta_\xi = 1$, $T_o = T_\xi = 10$, and $k_{to} = k_{t\xi} = 0.1$. The gain of the first QP solver in [17] is 100, and the second is 50.

As with the simulations, there are three cases in this subsection: HQP-PPC-AO2, HQP-PPC-AO1, and HQP-PPC-NAO. Fig. 11 shows the trajectory of the WMMR in these three cases. It can be seen that two obstacles collide when the robot

TABLE I
MAIN VARIABLES ADOPTED IN THIS ARTICLE

n	The number of control inputs.
m	The DOF of WMMR. $m = 3 + m_a$, where m_a is the DOF of manipulator.
$O_e \in R^3$	O_e is the end-effector coordinates of WMMR.
$q \in R^m$	$q = [q_v^T, q_a^T]^T$ represents the position of WMMR, and $q_v = [x_v, y_v, \theta_v]^T$ represents the coordinates of the mobile platform, $q_a = [q_{a1}, \dots, q_{am_a}]$ denotes the joint configuration of the manipulator
$\dot{\xi} \in R^n$	$\dot{\xi} = [\dot{\eta}^T, \dot{q}_a^T]^T$ represents the velocity of WMMR, and $\dot{\eta} = [v, w]^T$ denotes the linear and angular velocity of the moving platform, $\dot{q}_a = [\dot{q}_{a1}, \dots, \dot{q}_{am_a}]$ is the velocity of the manipulator joint.
$\tau \in R^n$	$\tau = [\tau_l, \tau_r, \tau_{q1}, \dots, \tau_{qm_a}]$ represents the control inputs of WMMR.
$e_o \in R^3$	$e_o = [e_{o1}, e_{o2}, e_{o3}]^T$ is the position error of the end effector.
$\hat{e}_o \in R^3$	$\hat{e}_o = [\hat{e}_{o1}, \hat{e}_{o2}, \hat{e}_{o3}]^T$ is the normalized position error of the end effector.
$\varepsilon_o \in R^3$	$\varepsilon_o = [\varepsilon_{o1}, \varepsilon_{o2}, \varepsilon_{o3}]^T$ is the transformed position error of the end effector.
$\rho_o \in R^3$	$\rho_o = [\rho_{o1}, \rho_{o2}, \rho_{o3}]^T$ is the bound of position error.
$e_\xi \in R^n$	$e_\xi = [e_{\xi1}, \dots, e_{\xi n}]^T$ is the velocity error in configuration space.
$\hat{e}_\xi \in R^n$	$\hat{e}_\xi = [\hat{e}_{\xi1}, \dots, \hat{e}_{\xi n}]^T$ is the normalized velocity error in configuration space.
$\varepsilon_\xi \in R^n$	$\varepsilon_\xi = [\varepsilon_{\xi1}, \dots, \varepsilon_{\xi n}]^T$ is the transformed velocity error in configuration space.
$\rho_\xi \in R^n$	$\rho_\xi = [\rho_{\xi1}, \dots, \rho_{\xi n}]^T$ is the bound of velocity error.
$\dot{\xi}_r \in R^n$	$\dot{\xi}_r = [\dot{\xi}_{r1}, \dots, \dot{\xi}_{rn}]^T$ is the reference velocity obtained by HQP.
$\dot{\xi}_s \in R^n$	$\dot{\xi}_s = [\dot{\xi}_{s1}, \dots, \dot{\xi}_{sn}]^T$ is the velocity obtained by the second-level QP that is used to guide the direction of obstacle avoidance.
$\bar{\Omega}_o$	$\bar{\Omega}_o$ is the set of all ε_{oi} that satisfy $ \varepsilon_{oi} \leq \bar{\varepsilon}_{oi}, i = 1, 2, 3$.
$\bar{\Omega}_\xi$	$\bar{\Omega}_\xi$ is the set of all $\varepsilon_{\xi i}$ that satisfy $ \varepsilon_{\xi i} \leq \bar{\varepsilon}_{\xi i}, i = 1, \dots, n$.

turns OFF the obstacle avoidance function, and each obstacle is successfully avoided when the obstacle avoidance function is turned ON. The experimental results of HQP-PPC-AO2 are shown in Fig. 12. Fig. 12(a) shows the velocity, which verifies the superiority of the proposed method in implementing the physical constraints of the robot. Fig. 12(d) shows the tracking errors, which reflects the high accuracy of our method. Fig. 12(b) shows the joint angles of the manipulator. Fig. 12(c) shows the control input.

B. Test of the HQP Function

In this subsection, we set the motors to velocity mode and used QP to determine the reference velocity.

Fig. 13 shows the WMMR velocity and the reference velocity planned through QP. The reference velocity $\dot{\xi}_r$ varies greatly. Although the reference velocity satisfies the velocity constraint, the actual velocity violates it. As shown in Fig. 16, the robot successfully avoided the first obstacle but collided with the second obstacle.

C. Test of the PPC Function

In this subsection, similar to the simulation, ARC and FLC were used instead of PPC. Figs. 14 and 15 show the velocity and tracking error of ARC and FLC, respectively. Compared with Fig. 12(a) and (d), PPC has better tracking accuracy and meets the velocity constraints.

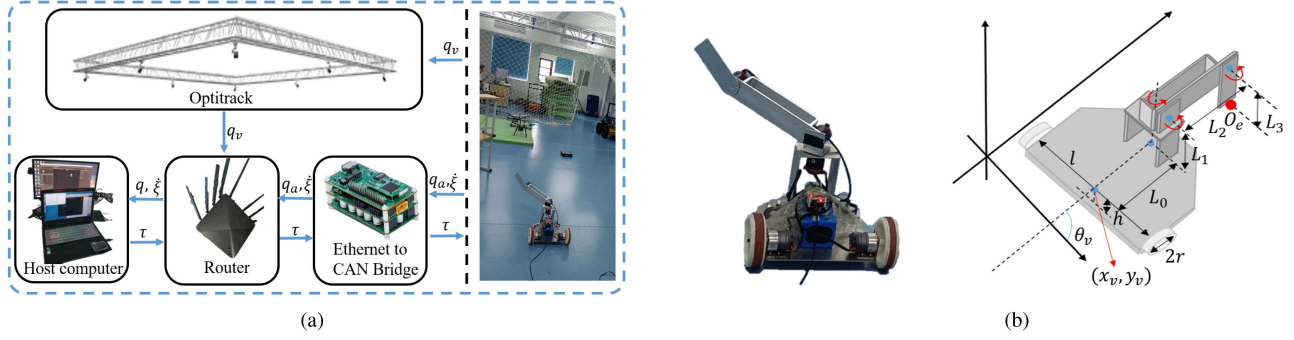


Fig. 10. Experimental facilities. (a) Control system of the WMMR. (b) Mechanical structure of the two-axis mobile manipulator.

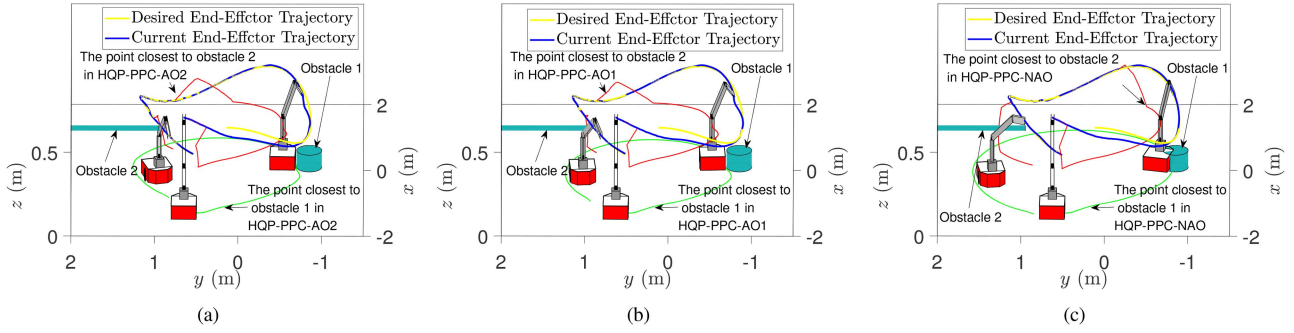


Fig. 11. Experimental trajectory of the WMMR by using HQP-PPC. (a) HQP-PPC-AO2. (b) HQP-PPC-AO1. (c) HQP-PPC-NAO.

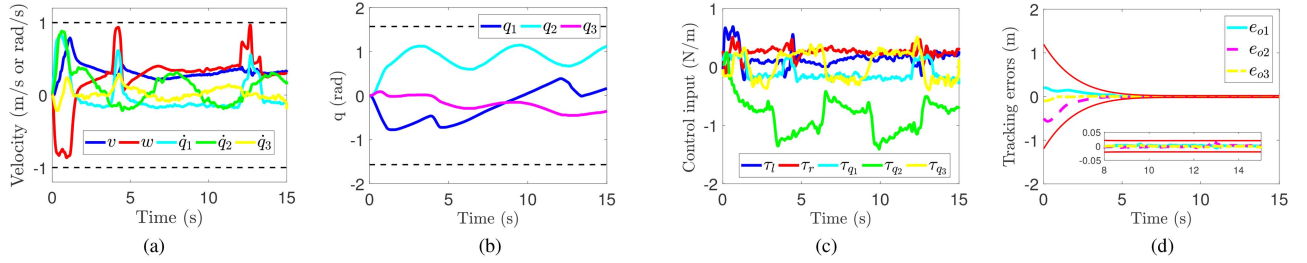


Fig. 12. Experimental results of HQP-PPC-AO2. (a) Velocity of the WMMR. (b) Joint angles of the manipulator. (c) Control input. (d) Tracking errors.

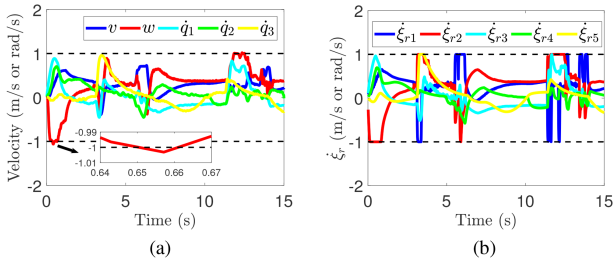


Fig. 13. Experimental results of QP. (a) Velocity of the WMMR. (b) Reference velocity ξ_r by solving QP.

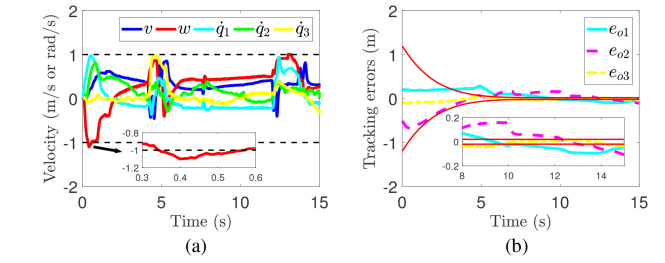


Fig. 14. Experimental results of ARC. (a) Velocity of the WMMR. (b) Tracking errors.

VI. DISCUSSION

The results of simulations and experiments show that several control objectives of this method have been achieved, which are as follows:

- 1) the tracking accuracy of the end effector is guaranteed;
- 2) all joint angles and velocity constraints are realized;
- 3) obstacle avoidance is realized.

To achieve the control objectives simultaneously, the whole planning-control scheme must also have the following features.

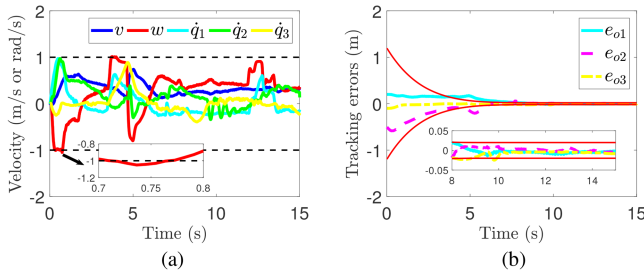


Fig. 15. Experiment results of FLC. (a) Velocity of the WMMR. (b) Tracking errors.

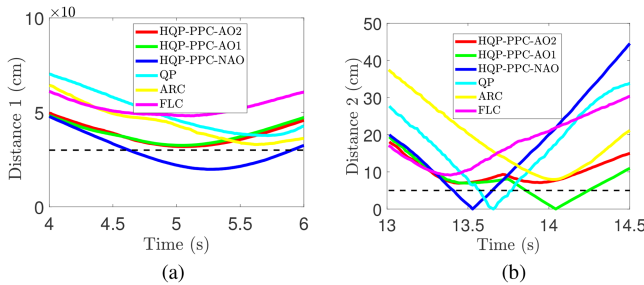


Fig. 16. Distances between the robot and the obstacle in the experiment. (a) Distance 1. (b) Distance 2.

- 1) The scheme should provide a smooth reference trajectory. In this article, to avoid producing an untraceable reference trajectory, the obstacle avoidance function is improved, and the obstacle avoidance direction is determined by HQP.
- 2) The controller can ensure tracking accuracy and predetermine the tracking error.

The characteristics of the controller can provide the planner with a known tracking error to obtain a reasonable trajectory and ensure a better execution of the planned trajectory.

To make a controller with excellent characteristics for which the tracking performance can be predetermined theoretically, the BLF and PPF are used in [28] and [30]–[33]; however, when approaching the boundary, these methods are at risk of chattering and boundary breaching. These risks can be reduced in some ways. On the one hand, increasing the control frequency as much as possible can significantly reduce the robot's motion in one sampling time, thus decreasing the probability of the WMMR approaching the boundary. On the other hand, reasonable control parameters can also avoid risks to a certain extent. These control parameters should be rectified several times in the experiment to achieve the optimal control effect.

In addition to the methods mentioned and compared in this article, there are other commonly used planning methods, such as kinodynamic probabilistic motion planning. This planning method uses heuristic algorithms on random discrete way points, taking into account the dynamic constraints of the robot, to reach the target position by node expansion and pruning of erroneous nodes. In general, this method is usually applied to point stabilization problems, and due to its predictive ability, a sufficiently good reference trajectory can be provided in a

complex environment. In contrast, it requires higher computing power. Although this approach is seldom applied to the trajectory tracking problem, the idea of using the probability method to produce an optimal trajectory may provide a new method for redundant mobile manipulators.

VII. CONCLUSION

In this article, an HQP-PPF control scheme was proposed to realize the end-effector tracking, obstacle avoidance, and physical constraints of the WMMR. At the kinematic level, the method used HQP to plan the movement in the configuration space. The first-layer QP was used to avoid obstacles and realize physical constraints, and a smoother obstacle avoidance function was proposed. The second-layer QP was used to guide the direction of obstacle avoidance. On the other hand, the PPF can not only ensure the tracking accuracy of the end effector but also accurately realize the planned trajectory so that the actual movement of the WMMR can achieve obstacle avoidance and cannot violate the constraints. Finally, the effectiveness of the proposed method was verified by simulations and experiments.

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